Zaboraa FX

Requirements Specification and Project Status

Version 4.3 | Updated 27 Oct 2026 | Supersedes v4.2 and every earlier version.

Item	Content
Purpose	Master brief and living status record: what the product must do, what is built, what is verified, what is pending.
Scope of this edition	v4.3 carries every section that changed since v4.2: what changed, status at a glance, progress log, next actions, the acceptance tracker (new), requirement rows, the defect list D1 to D42, the manual checklist and the open decisions. Sections 3 to 7, 10 to 13, 16 and 17 of v4.0 are not repeated here and are unchanged except where the master plan v4.3 says so.
Basis of the status	The owner's chat of 27 Oct 2026: the EURUSDm download failures fixed, the Active Research Dataset issue fixed, and the supplied project status notes. The editor read the uploaded server.ts (it contains the symbol-suffix and ISO date changes) and the uploaded main.py (the pre-fix copy, so the bridge paging fix was not seen). Nothing here was executed by the author of this document; no test names were cited for the 27 Oct fixes.
Evidence labels	VERIFIED = a test or recorded run is cited and the code was seen. OWNER-REPORTED = the owner says they ran it. CLOSED IN CODE AND TESTS = fix and named tests reported, live check still open. GATE-PARTIAL = tests pass, something named is open. OPEN = not done. PROPOSED = found by the editor, not reported by the owner.
Disclaimer	Research and automation tool, not financial advice. Past backtest performance does not guarantee future results. Trading carries risk of loss.

Read this first. On 27 Oct the owner reports two blockers cleared: the historical download now works past page 1 for EURUSDm (broker-suffix mapping and bridge paging, D38 and D39) and stored candles now become active research datasets (D40). Four things keep the status honest. (1) The uploaded main.py is still the pre-fix copy, so the bridge fix is reported, not seen, and has no named test. (2) Only EURUSDm M1 paging is reported; all 8 timeframes, the other symbols, a repeat download that adds nothing, page-boundary integrity and restart persistence of the research dataset and its provenance are not yet shown. (3) The last recorded gate is an undated 260/260 and pytest 61; nothing was re-run after these fixes, so the current project has no recorded gate. (4) The D29 to D32 live checks, D36 and D37 are unchanged and open, so ALLOW_TRADING=false and ALLOW_LIVE=false stay in force. Overall: advanced development, moving into integration validation; not certified for unattended automation or live trading. Rounds 2, 3 and 4 stay GATE-PARTIAL.

1. What changed

New in v4.3 (this version)

- **Broker symbol mapping, D38 (OWNER-REPORTED, seen in server.ts).** The portal sent EURUSD; the Exness demo terminal only knows EURUSDm, so /candles answered 404. /api/data/bulk-download now appends brokerState.symbolSuffix (default m) when it is missing, calls the bridge with the broker symbol and stores candles under the canonical symbol; the job record keeps both. Another route (normalizedSymbol) applies the same rule; confirm one resolver is used everywhere.
- **Pagination date format (seen in server.ts).** The page loop sends date_to as ISO-8601 with +00:00 instead of epoch seconds; the bridge no longer answers 400 INVALID_DATE_RANGE.
- **Bridge paging bug, D39 (OWNER-REPORTED fixed; main.py not seen).** Page 2 then failed with 502 (-2, 'Terminal: Invalid params'). Cause: with date_to and no date_from the bridge ran copy_rates_range from 2000-01-01, a 26-year range. Fix: with only date_to call copy_rates_from(symbol, tf, end_utc_aware, count); copy_rates_range only when date_from is given; page 1 keeps copy_rates_from_pos from position 1 so the forming bar stays excluded. Candle time stays broker-time epoch and date_to round-trips in one basis, so no offset conversion belongs here.
- **Diagnosis correction.** The earlier explanation (naive-UTC datetimes and URL-decoding with unquote) was not the cause and is withdrawn. FastAPI already decodes query strings. Any notes that list naive-UTC conversion as a fix for copy_rates_range should be corrected.
- **Active Research Dataset, D40 (OWNER-CONFIRMED).** Stored candles did not become active research datasets, so Backtesting Lab reported zero and blocked simulations. The owner replaced the files from the supplied ZIP and reports it fixed; the files were not named or shown. Open: restart persistence, persisted provenance (the 4.4 summary says it lived only in memory), and canonical-symbol matching of XAUUSD with XAUUSDm.
- **Found by reading server.ts (PROPOSED, not tested).** D41: the bridge answers an empty result with 502 NO_DATA and the page loop treats every non-OK reply as failure, so a timeframe whose history ends exactly on a full page fails after saving its data. D42: the loop stops at 200 pages of 5,000 bars (1,000,000 bars per timeframe per job), about 2.7 years of M1 at the plan's 370,000 bars a year, and a repeat job restarts from the newest bar.
- **Gate figures (OWNER-REPORTED, undated).** An earlier project version is reported as 260 of 260 tests with lint and the production build clean (v4.2 had '257+'). The 4.4 implementation summary records bridge pytest 61 passed and a 64-file syntax transpile, with the full type-check, Vitest and Vite build not run in its environment. Nothing was re-run for v4.3.
- **Unchanged and still OPEN:** D29 to D32 live checks (checklist 8, 21, 27, weekday offset), D18, D20, D21, D22 (rotate BRIDGE_TOKEN and ENCRYPTION_MASTER_KEY), D25, D26, D16, D36, D37.

Earlier versions (summary). v4.2: Vertex AI Pine conversion, comparator and backtester hardening, D19 and D35 closed in code, D28 partly wired, D36 and D37 added, '257+' tests. v4.1: pytest 61/61, vitest 253/253, D29 to D33 closed in code and tests with live checks open, D34 closed, D35 added. v4.0: vitest 194/194 in 12 files, D27 closed, D28 partial, D29 to D34

found. v3.9 to v3.8: Rounds 1 to 3 built; Bearer-only auth; 8 timeframes; data routes. v3.7 and earlier: 119/119, runner ordering, BackgroundJob, laptop-only decision, backtest track, selection protocol.

2. Status at a glance (27 October 2026)

Area	Status	Summary
MT5 bridge (Python)	CLOSED IN CODE AND TESTS; live checks OPEN; paging fix OWNER-REPORTED	pytest 61/61 last recorded. Real /kill, order path, UNKNOWN account type, no synthetic candles, symbol hardening. Live demo-position, order-reject, terminal-closed and weekday offset checks pending (D29 to D32). Paging fix (D39) reported, main.py not seen, no test for the date_to branch.
Historical download (D38, D39)	FIXED, owner-reported (EURUSDm M1)	Suffix mapping and paging reported working. Open: all 8 timeframes, other symbols, repeat download adds 0, page boundaries, end of history (D41), depth cap (D42).
Active Research Dataset (D40)	FIXED, owner-confirmed	Backtesting Lab sees active datasets. Open: restart persistence, persisted provenance, XAUUSD and XAUUSDm matching, named tests.
Safety foundation (Stages 0-2B)	DONE	QuickJS sandbox, encrypted credentials, master-key refusal, API host/origin firewall, mandatory stop-loss, kill-switch server logic, no-lookahead guard. Unchanged.
Stage D1 security	GATE-PARTIAL	apiSecurityMiddleware, token masking, HTML guards, account-type check before idempotency log, Bearer-only client. D25 open.
Order path and reconciliation (3A)	DONE against a mocked bridge	Not yet exercised with a real demo order (D30 live check).
Bots, risk manager, watchdog (3B)	GATE-PARTIAL	D19 spread filter reported fixed (named test to confirm). 3B audit table unfilled.
Fake-data removal (4)	GATE-PARTIAL	D35 reported fixed (named test to confirm). Modify-position still local (D21). Synthetic startup datasets removed (4.4 summary).
Backtest, optimization, Monte Carlo	BUILT	20% final holdout reserved in the optimizer and strict candle validation (4.4 summary). Walk-forward, a held-out guard in code (FR-N3), PDF reports and multi-symbol missing.
Backtest engine and comparator (Round 4)	GATE-PARTIAL	Comparator and backtester hardened; 4-test validation file (owner-reported). Not yet compared with real TradingView exports.
Pine conversion	BUILT; Vertex AI since 26 Oct	Manual conversion works (owner-reported). D36 and D37 open. 3-sample TradingView comparison pending.
Rounds 2 and 3 (data layer)	GATE-PARTIAL	Server side tested; UI wiring reported 26 Oct. Open: resume, retry, earliest bar, data-exists UI, Market closed label, weekday acceptance, weekend gap test on real data.
Local verification gate	NOT RECORDED for the current project	Last reported: 260/260 (undated), pytest 61. Re-run npm ci, lint, test, npx tsc --noEmit, build and pytest and record the output.
Deployment (Vercel)	OPEN (D16)	Owner check of project and repo privacy. Vertex ADC is local-only; hosting elsewhere would need a service account (Stage 10).
Live trading	PENDING	Disabled. Needs the live checks for D29 to D32, wiring gate, manual checklist, 1+ week demo soak, ALLOW_LIVE and typed confirmation.

2A. Progress log (new and changed entries)

Item	Result	Status
Symbol suffix in bulk download (27 Oct)	EURUSD sent to the bridge as EURUSDm; stored as EURUSD. 404 gone.	OWNER-REPORTED; seen in server.ts
ISO date_to for paging (27 Oct)	+00:00 offset string; 400 gone.	OWNER-REPORTED; seen in server.ts
Bridge date_to branch (27 Oct)	copy_rates_from with count replaces the 2000-01-01 range; 502 Invalid params gone for EURUSDm M1.	OWNER-REPORTED; main.py not seen; no named test
Active Research Dataset (27 Oct)	Backtesting Lab sees active datasets after the supplied files replaced the old ones.	OWNER-CONFIRMED; files not named
Restart persistence and provenance of research datasets	Not checked.	OPEN
Eight-timeframe download, EURUSDm and XAUUSDm; repeat adds 0; page-boundary audit	Not done.	OPEN
End of history (D41) and depth cap (D42)	Found in code reading; no run.	PROPOSED
Full local gate on the current project	Not run since the 27 Oct fixes.	OPEN
Browser end-to-end convert test, D36	Not reported.	OPEN
D19 and D35 named tests	Not cited.	CLOSED IN CODE, NAMED TEST TO CONFIRM
Position-close check, checklist 21	Not done: needs a demo position.	OPEN
Order rejected with ALLOW_TRADING=false, checklist 8	Not done.	OPEN

Item	Result	Status
Terminal closed: /account and /candles, checklist 27	Not done.	OPEN
Weekday UTC offset measurement (D32)	Not done.	OPEN
Round 2 weekday acceptance, checklist 23	Not done.	OPEN
Code recheck of changed files	Not done: main.py (fixed copy), test_bridge.py, botRunner.ts, backtester.ts, the dataset-activation files and the new tests have not been seen in full.	OPEN
Rotate keys (BRIDGE_TOKEN, ENCRYPTION_MASTER_KEY), Vercel and repo privacy, commit on local-verified, Payop question	Values were pasted into a chat on 26 Oct; rotate (D22). Others unchanged from v4.0.	OPEN (D22, D16)

Next actions, in order

- Back up the project and data/zaboraa.sqlite. Run npm ci, npm run lint, npm test, npx tsc --noEmit, npm run build and python -m pytest test_bridge.py -v on the current integrated project; record the exact output.
- Send the current zip (no .env, no Google credential files) and the mt5-bridge folder so the fixed main.py and the dataset-activation files can be rechecked in code. Name the tests behind D19, D35 and the 27 Oct fixes.
- Add tests: bridge pytest for the date_to branch (copy_rates_from, tz-aware UTC, requested count, bad date rejected); server tests for the symbol suffix with and without it and for a symbol the broker lacks.
- Data acceptance on the Exness demo (checklist 0e): EURUSDm and XAUUSDm, all 8 timeframes, repeat download adds 0 bars, no gap or overlap at page boundaries, newest stored bar older than the forming bar.
- Restart the portal and confirm the research dataset reloads with persisted provenance and that XAUUSD finds XAUUSDm data; add a restart test (D40).
- D41: treat 502 NO_DATA after at least one saved page as the end of history. D42: measure the broker's real M1 depth before changing the 200-page cap.
- D36: confirm type="button" and e.preventDefault() are saved; run the browser convert test. D37: update .env.example and docs; one Vertex client for every Gemini call; failing-case test.
- Run the live demo checks: demo position then /kill and /positions (21); order with ALLOW_TRADING=false (8); terminal closed, /account and /candles (27); weekday UTC offset (D32).
- Rotate BRIDGE_TOKEN (portal and bridge) and ENCRYPTION_MASTER_KEY (re-enter stored credentials), see D22.
- Finish D28, then Round 4: tighten the integration test, real 3-sample Pine comparison, held-out guard (FR-N3), walk-forward. Close D25 and D26; commit on branch local-verified once the gate passes.

3B. Acceptance tracker (new in v4.3)

A blank cell means no evidence is on record. Update this table at every checkpoint.

Requirement	Implemented	Named test	Owner-confirmed live	Evidence still missing
Broker-symbol mapping (D38)	Yes, seen in server.ts	None cited	No more 404 (EURUSDm)	Test with/without suffix; other symbols; one resolver
Bridge paging with date_to (D39)	Reported; not seen	None cited	Page-2 error gone (EURUSDm M1)	Recheck main.py; date_to pytest; boundary audit
End of history (D41)	No			Implement and test both cases
Download depth (D42)	200-page cap			Measure M1 depth; decide on extend mode
8 timeframes download and store (FR-H10)	Yes (4.4)	dataApi.test.ts (server side)		All 8 on the demo terminal, twice
Append-only, forming candle excluded	Yes (4.4)			Repeat adds 0; newest bar older than current
Active Research Dataset (D40)	Files replaced	None cited	Datasets visible	Restart, provenance, symbol matching, tests
Backtest validation, 20% holdout	Yes (4.4)	backtestValidation.test.ts		Run on current build; FR-N3 guard
Reports, markers, print to PDF	Yes (4.4)			Browser run: save, reopen, delete, PDF
Engine versus TradingView	Comparator built	Well-formed output only		Three real Pine samples
Pine on Vertex AI (D36, D37)	Yes		Manual conversion	Browser test; failing-case route test
Bridge safety live checks (D29 to D32)	Closed in code	pytest 61	Sunday empty /kill only	Checklist 8, 21, 27; weekday offset
Local gate		260/260 (undated); pytest 61		Fresh full run on the current project

Requirement	Implemented	Named test	Owner-confirmed live	Evidence still missing
Stage B, W, S, L entry criteria	See master plan			B: rounds 4 to 6. W: audit table. S: W plus a one-week demo. L: all, plus explicit owner approval

8. Requirement rows changed in v4.3

ID	Requirement	Status	Notes
FR-H11 (new)	Download uses the broker-suffixed symbol and stores the canonical symbol	GATE-PARTIAL	Seen in server.ts (D38); named test and all-symbol check open.
FR-H12 (new)	A page is count bars ending at date_to; end of history is a normal completion	PARTIAL	Paging fix reported (D39); end-of-history handling open (D41).
FR-N7 (new)	Genuine MT5 candles become active research datasets, also after a restart; provenance persisted	PARTIAL	Activation fixed (D40, owner-confirmed); restart and provenance open.
FR-H1	Bulk download of all timeframes	GATE-PARTIAL	Paging now works for EURUSDm M1 (owner-reported); all 8 timeframes, resume, retry, earliest bar open (D28, D41, D42).
FR-N3	Held-out data rule enforced in code	OPEN	4.4 summary: optimizer reserves the final 20% and excludes it from scoring and ranking; a guard that stops all ranking and walk-forward paths from reading held-out bars is not shown.
FR-D2, D3, T5, H9; NFR-S2, W1; FR-N6; Bot spread filter; FR-B9; FR-M2, N5, H3, H10		Unchanged from v4.2	See v4.2 notes; D28 now also depends on the 8-timeframe acceptance.

The three new IDs are the editor's wording of what the fixes imply; the owner accepts, edits or drops them.

9. Known defects

ID	Defect	Status (27 Oct, v4.3)
D1-D8	Round 0: bot runner, watchdog feed, client-side fakes, kill-modal, HOST, browser token, static prices, README	GATE-PARTIAL (code recheck open); D7 and D8 partial
D9-D11	Status docs, Stage 4 tests, interrupted 3B run	D9 PARTIAL; D10 and D11 OPEN
D12, D14, D15, D23	Old test count, token shared, browser offline, invalid-symbol tick test	RESOLVED (D23 confirm in code)
D13, D17	main.py edits; few bridge tests	PARTIAL: pytest 61; confirm CORS, demo detection, order-rejection cases; add the date_to test
D16	Public Vercel deployment; repo privacy	OPEN (owner)
D18	queryBridgeAccountType assumes DEMO when unreachable	PARTIAL: bridge side fixed (D31); confirm portal fails closed
D19	Bot spread filter receives no spread value	CLOSED IN CODE (owner-reported); named test to confirm
D20	/tick time field assumed	OPEN: confirm field name; stale and fresh fixtures
D21	Modify-position local-only	OPEN
D22	Uploaded zip contained a real .env; ENCRYPTION_MASTER_KEY and BRIDGE_TOKEN values also pasted into a chat on 26 Oct	OPEN (owner): rotate BRIDGE_TOKEN in portal and bridge; a new master key makes stored credentials unreadable, so re-enter them
D24	Bearer-only bridge auth	RESOLVED? (owner-reported); confirm all callers
D25	Preview-domain switch widens loopback rule	OPEN
D26	Ephemeral encryption key allowed in tests	OPEN
D27	Round 2 and 3 routes had no named tests	CLOSED (v4.0)
D28	Round 2 and 3 items not reported	PARTIAL (v4.3): chart and Data Viewer wiring, market indicator, cancel, M1 warning reported 26 Oct; the paging failure that blocked the acceptance (D38, D39) reported fixed 27 Oct. Open: resume, retry, earliest bar, data-exists UI, Market closed label, weekday acceptance on all 8 timeframes, weekend gap test
D29	/kill reported flat without closing anything (safety critical)	CLOSED IN CODE AND TESTS. LIVE CHECK OPEN (checklist 21)
D30	POST /orders and delete returned mock tickets	CLOSED IN CODE AND TESTS. LIVE CHECK OPEN (checklist 8)
D31	Bridge reported DEMO when MT5 disconnected	CLOSED IN CODE AND TESTS (now UNKNOWN). LIVE CHECK OPEN (checklist 27)

ID	Defect	Status (27 Oct, v4.3)
D32	utc_offset_hours hard-coded to 0	CLOSED IN CODE AND TESTS. LIVE MEASUREMENT OPEN (weekday)
D33	Synthetic candles returned when MT5 down	CLOSED IN CODE AND TESTS; portal guard kept
D34	Symbol regex accepts . and ..; no length cap	CLOSED AND VERIFIED BY TESTS (owner-reported); confirm length cap in code
D35	MarketsView shows a false Order executed message	CLOSED IN CODE (owner-reported); named test to confirm
D36	Buttons without type="button" trigger implicit form submits and page reloads	OPEN: fix provided; applied and browser-tested not yet reported
D37	Vertex AI migration not reflected in .env.example and docs; other Gemini routes may still use the key; no failing-case test; credits can expire; model id may retire	OPEN
D38 (new)	Bulk download sent canonical symbols; the Exness demo terminal needs suffixed names, so /candles answered 404	CLOSED IN CODE (owner-reported; seen in server.ts); named test and all-symbol acceptance to confirm; one resolver for every route
D39 (new)	Page 2 failed with 502 (-2, Terminal: Invalid params): with date_to and no date_from the bridge called copy_rates_range from 2000-01-01. The earlier tz-aware and unquote explanation was wrong	CLOSED (owner-reported). main.py fix not seen; no named test; scope tested: EURUSDm M1
D40 (new)	Stored candles did not become active research datasets; provenance only in memory; exact-symbol filtering can hide XAUUSDm from an XAUUSD context	ACTIVATION CLOSED (owner-confirmed). OPEN: restart persistence, persisted provenance, canonical-symbol matching, named tests
D41 (new)	The page loop treats the bridge's 502 NO_DATA as failure, so history ending exactly on a full page fails the job after saving	OPEN (proposed; from reading server.ts)
D42 (new)	The loop stops at 200 pages of 5,000 bars (1,000,000 per timeframe per job, about 2.7 years of M1) and every job restarts from the newest bar	OPEN (proposed; derived, not tested; measure first)

14. Manual MT5 checklist: rows changed in v4.3

#	Step	Status
5	python -m pytest test_bridge.py -v: all pass	OWNER-REPORTED (61 passed, 25 Oct); re-record; add the date_to case
8	ALLOW_TRADING=false: send an order through the bridge; rejected, no mock ticket	OPEN (code and tests closed, D30)
14	Gate: npm install, lint, test, build; counts in 2A	OPEN for the current project: last reported 260/260 (undated), tsc 0, build clean; re-record everything
21	Demo position open: /kill then /positions; position really closed, flat reported only after re-read	OPEN (code and tests closed, D29; Sunday empty case only)
27	MT5 closed: /account not DEMO, /candles JSON error	OPEN (code and tests closed, D31, D33)
28	/api/data/load and /audit with .. and .: rejected before disk access	Tests pass (owner-reported); run once on the laptop
29	Pine convert in the browser: Convert stays on the Pine view, code appears, Save opens Strategy Studio; gcloud ADC still valid	OPEN (D36)
0e (new)	Download acceptance, Exness demo, weekday: bridge log shows EURUSDm and 200 on every page; repeat adds 0 bars and no duplicate timestamps; audit shows no gap or overlap at page boundaries; newest stored bar older than the forming bar; repeat for XAUUSD and all 8 timeframes; after a portal restart the dataset appears in Backtesting Lab as eligible and XAUUSD finds XAUUSDm; record pages, bars and earliest bar per timeframe	OPEN
Others 1-4, 6-7, 9-13, 15-20, 22-26		Unchanged from v4.0

15. Open decisions

Unchanged from v4.0, except that the broker UTC offset fallback should be "measure; if it fails, mark the dataset offset unknown" (v4.1), that the AI backend is Vertex AI with local Application Default Credentials (v4.2), and, new in v4.3: (a) the owner decides whether to extend M1 depth beyond about 2.7 years per job (D42) after the broker's real depth is measured; (b) the owner accepts, edits or drops FR-H11, FR-H12 and FR-N7. Keep the model id only in GEMINI_MODEL and keep ALLOW_TRADING=false until the live checks pass.

Document history: v1.0 (23 Oct) brief. v2.0 to v3.9 (24 to 25 Oct) architecture, rounds, hardening, Rounds 1 to 3. v4.0 (25 Oct) vitest 194/194, D27 closed, D29 to D34. v4.1 (25 Oct): pytest 61/61, vitest 253/253, tsc 0, lint and build clean; D29 to D33 closed in code and tests with live checks open; D34 closed; D35 added. v4.2 (26 Oct): Vertex AI backend, Pine conversion working in a manual test, comparator and backtester hardening, D19 and D35 closed in code, D28 partly wired, D36 and D37 added, '257+' tests, tsc 0, build clean. v4.3 (27 Oct): symbol suffix and ISO paging dates in bulk download (D38), bridge paging fix (D39), Active Research Dataset activation fixed (D40), D41 and

D42 proposed, acceptance tracker and checklist 0e added, gate recorded as 260/260 (undated) and not re-run.